University of BRISTOL

COMS20018, Introduction to AI: Causal Inference

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Agenda

- Motivation
- Potential outcomes
- Identification assumptions
- Causal graphs
- The do-operator
- Identification

Motivating example: Simpson's paradox

A (hypothetical) new disease, **COVID-X**, has two treatments: **Treatment 0** (common) and **Treatment 1** (scarce). You must pick *one* for the whole country, to **minimise deaths**.

Death rate	Total
Treatment 0	**16%** (240/1500)
Treatment 1	19% (105/550)

Lower is better. Counts are deaths / patients.

See Neal (2020), Section 1.1, Table 1.1.

Motivating example: Simpson's paradox

A (hypothetical) new disease, **COVID-X**, has two treatments: **Treatment 0** (common) and **Treatment 1** (scarce). You must pick *one* for the whole country, to **minimise deaths**.

Death rate	Mild	Severe	Total
Treatment 0	15% (210/1400)	30% (30/100)	**16%** (240/1500)
Treatment 1	**10%** (5/50)	**20%** (100/500)	19% (105/550)

Lower is better. Counts are deaths / patients.

Question

The totals favour Treatment 0, but within each condition Treatment 1 has the lower death rate. Which treatment would you choose for the country, and why?

See Neal (2020), Section 1.1, Table 1.1.

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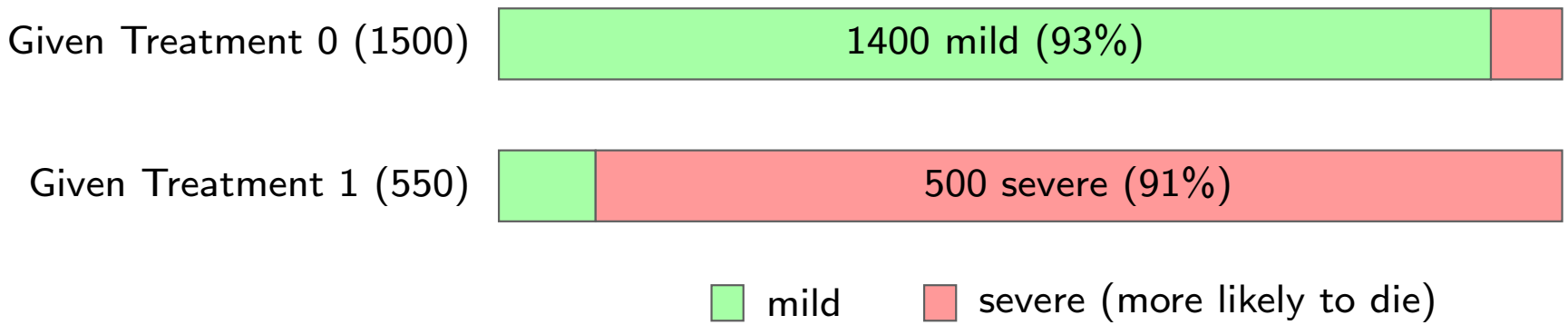
Lower is better. Counts are deaths / patients.

Simpson's paradox: Treatment 0 wins *overall*, but Treatment 1 wins in *both* the mild *and* the severe groups. Which do we believe?

See Neal (2020), Section 1.1, Table 1.1.

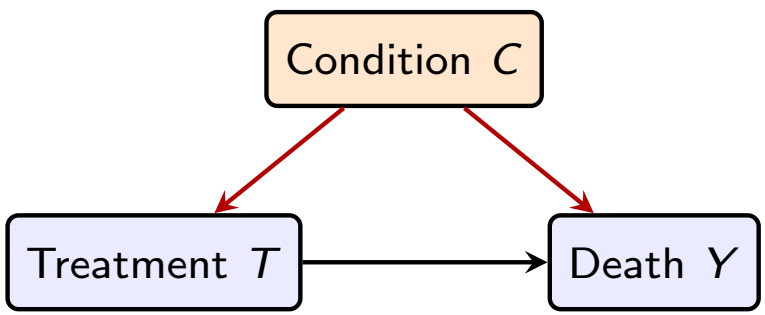
The answer depends on the causal relationship

Who received which treatment?



Doctors gave common Treatment 0 to most **mild** cases and saved scarce Treatment 1 for **severe** ones.

Condition C affects *both* which treatment T a patient gets *and* whether they die (Y): a **common cause**.



See Neal (2020), Section 1.1, Figure 1.1.

Prediction $\neq$ causation

Great for prediction

- Predict patient deaths
- Rank search results
- Classify images
- Complete the next token

No causal claim needed: a better fit is simply better.

vs

Needs causation

- Would Treatment 1 reduce deaths?
- Does the ad cause the purchase?
- Does this prompt cause bias?
- What would another training strategy give?

A better fit can still give the wrong causal answer.

$\Rightarrow$ Even a perfect predictor (zero generalisation error, infinite data) cannot answer causal questions: the gap is **structural**, not about sample size.

Part 1

Potential Outcomes

Formalising the question "what would have happened if. . ."

Potential outcomes | Assumptions | Causal graphs | do-operator | Identification

Formalising counterfactuals

Running example: COVID-X patients
$T \in \{0, 1\}$: the treatment received (Treatment 0 or Treatment 1);
$Y = 1$ if the patient dies, $Y = 0$ if they survive

What we observe

Patient	T	Y
1	0	0
2	1	1
3	1	0
4	0	0
5	0	1

Table adapted from Neal (2020), Section 2.1, Table 2.1.

Formalising counterfactuals

Running example: COVID-X patients
$T \in \{0, 1\}$: the treatment received (Treatment 0 or Treatment 1);
$Y = 1$ if the patient dies, $Y = 0$ if they survive

Potential outcomes

$Y(0)$ = outcome *if* Treatment 0 were used

$Y(1)$ = outcome *if* Treatment 1 were used

$Y(t)$ = potential outcome under treatment t

? = counterfactual (never observed)

Only one column per row is ever observed

Patient	T	$Y(0)$	$Y(1)$
1	0	0	?
2	1	?	1
3	1	?	0
4	0	0	?
5	0	1	?

Table adapted from Neal (2020), Section 2.1, Table 2.1.

Formalising counterfactuals

Running example: COVID-X patients
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Only one column per row is ever observed

Patient	T	$Y(0)$	$Y(1)$	ITE
1	0	0	?	?
2	1	?	1	?
3	1	?	0	?
4	0	0	?	?
5	0	1	?	?

$$\tau_i = Y_i(1) - Y_i(0) \quad \textbf{Individual Treatment Effect (ITE)}$$

$\Rightarrow$ $Y_i(1) - Y_i(0)$ is the effect of switching from treatment 0 to treatment 1 for patient i.

Table adapted from Neal (2020), Section 2.1, Table 2.1.

From individual to average treatment effects

Individual Treatment Effect (ITE)

$$\tau_i = Y_i(1) - Y_i(0)$$

Unobservable per unit: both columns are never seen simultaneously.

Average Treatment Effect (ATE)

$$\tau = \mathbb{E}[Y(1) - Y(0)]$$

Average across the population: the target of most causal inference methods.

Why not just use the difference in means?

$$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$$

Discuss with your neighbour (1 min).

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$$\mathbb{E}[Y \mid T{=}1] - \mathbb{E}[Y \mid T{=}0]$$

Compares **different populations**.

Patients given Treatment 1 were mostly severe; those given Treatment 0 mostly mild:

$T{=}0$: **93% mild** $T{=}1$: **91% severe**

This is **confounding**: the difference in means mixes the treatment effect with how sick the patients were.

Goal: identify when these two quantities are equal.

The fundamental problem of causal inference

It is impossible to observe both $Y(0)$ and $Y(1)$ for the same unit at the same time (Holland, 1986)

Factual

The potential outcome under the treatment that was *actually received*.



This is what appears in the data.

Counterfactual

The potential outcome under the treatment that was *not* received.



This is **never** in the data.

Causal inference is fundamentally a missing data problem.

See Neal (2020), Section 2.2.

Part 2

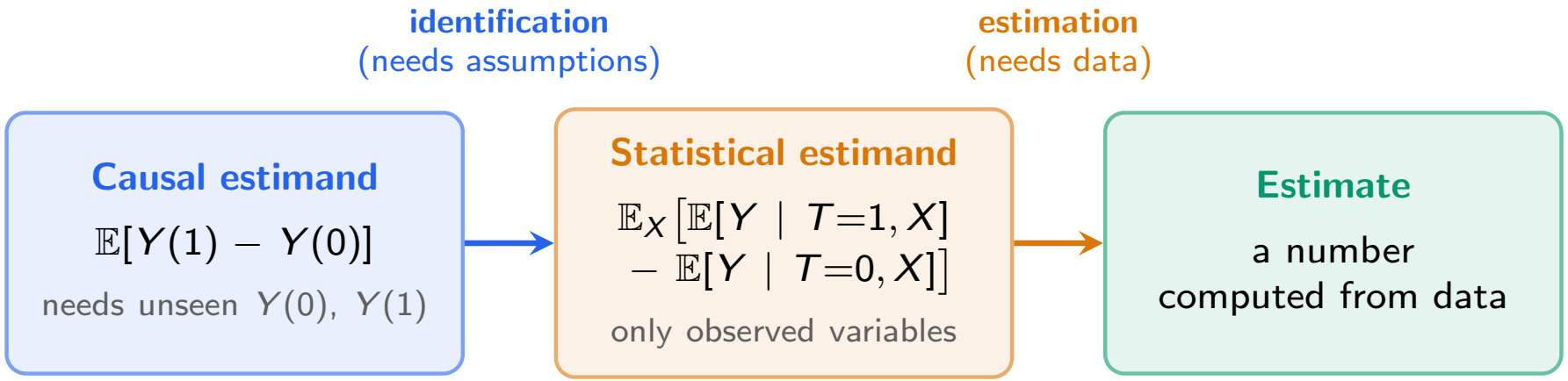
Identification Assumptions

When can we estimate causal effects from data?

Potential outcomes | **Assumptions** | Causal graphs | do-operator | Identification

What is causal identification?

- **Identification**: rewrite a causal quantity using only quantities we can estimate from observational data.



X: observed covariates (e.g. the patient's condition)

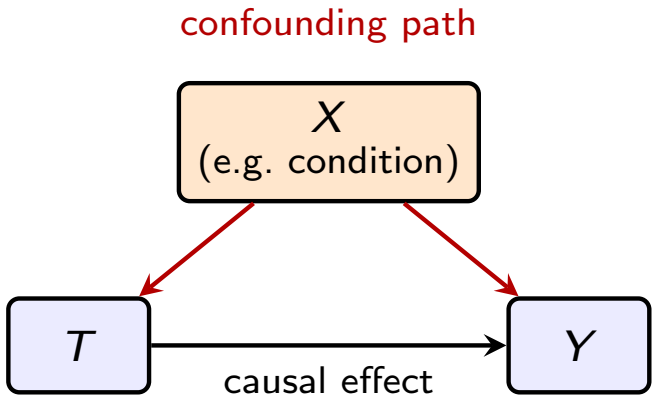
$\Rightarrow$ Identification needs **causal assumptions**, not more data.

See Neal (2020), Sections 2.3–2.4.

Assumption 1: unconfoundedness

$(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$ "conditional exchangeability" / "no hidden confounders"

- Within groups sharing the same covariates X, the potential outcomes are independent of which treatment was actually received (**no confounding**).



- **Violated when**: the treatment is chosen based on criteria that are **not recorded** in X and that also affect the outcome.

See Neal (2020), Section 2.3.

Assumption 2: positivity

$$0 < P(T = t \mid X = x) < 1 \quad \text{for all } x \text{ with } P(X = x) > 0$$

- Every sub-group must have positive probability of receiving any treatment value.

- **Violated when**: some group never receives some treatment, e.g. every severe patient received treatment 1:

$$P(T{=}0 \mid X{=}\text{severe}) = 0$$

Then $\mathbb{E}[Y \mid T{=}0, X{=}\text{severe}]$ is **undefined**: we have never observed a severe patient on treatment 0, so the effect for severe patients cannot be estimated.

See Neal (2020), Section 2.3.

Assumption 3: SUTVA

Stable Unit Treatment Value Assumption: two distinct conditions.

- **Consistency**: if $T = t$ then $Y = Y(t)$
  - The treatment received is exactly the treatment in the model.
  - **Violated when**: Treatment 1 is scarce, so some patients get it straight away and others only after a long wait, by which time their condition may have worsened.
  - **In ML**: distribution shift.
- **No interference**: $Y_i(t) \perp\!\!\!\perp T_j$ for $i \neq j$
  - One unit's outcome does not depend on another unit's treatment.
  - **Violated when**: treating patient j reduces spread, which changes patient i's outcome (an infectious disease).
  - **In ML**: exactly the i.i.d. assumption.

Three assumptions for causal identification

Under these three assumptions, the ATE is identifiable from observational data.

1 **Unconfoundedness** $(Y(0), Y(1)) \perp\!\!\!\perp T \mid X$

no hidden confounders

2 **Positivity** $0 < P(T{=}t \mid X{=}x) < 1$

every group can get every treatment

3 **SUTVA** consistency + no interference

well-defined treatment; units don't interact

Part 3

Causal Graphs

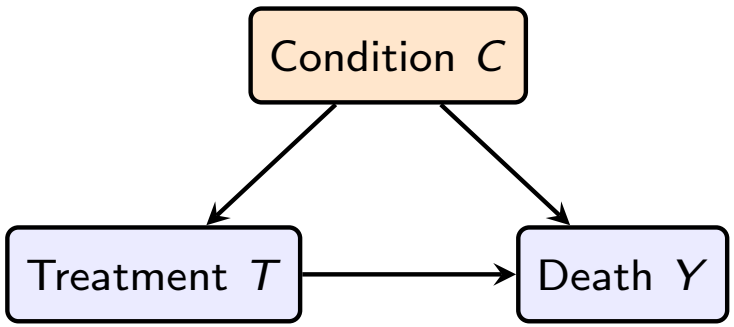
Reading the structure of cause and effect

Potential outcomes | Assumptions | **Causal graphs** | do-operator | Identification

Causal graphs: directed acyclic graphs (DAGs)

- **Nodes** = variables; **edges** = direct causal influence.
- **Directed**: $X \to Y$ means X is a direct cause of Y.
- **Acyclic**: no directed cycles; causes must precede their effects in time.
- **Markov**: each variable is conditionally independent of its non-descendants given its parents.

Example: COVID-X



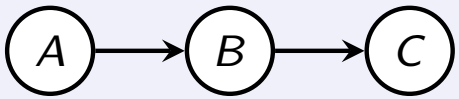
C is a parent of both T and Y

$\Rightarrow$ Same graph structure as a Bayesian network, but the edges claim **causal influence**, not just statistical dependence. This is what lets us reason about interventions.

See Neal (2020), Sections 3.1–3.3.

Three graphical building blocks

Chain



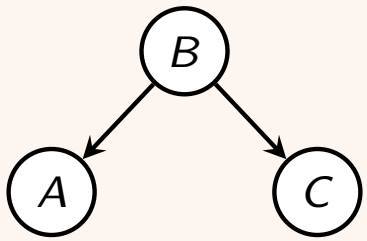
A, C associated?

Yes, through B

Condition on B: **blocks**

Treatment → condition worsens while waiting → death (mediator)

Fork



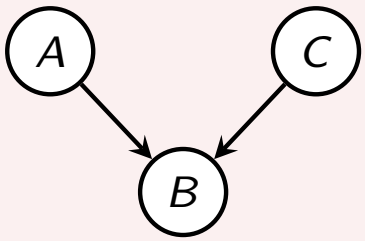
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Condition on B: **blocks**

Condition → treatment *and* death (confounding, as in COVID-X)

Collider



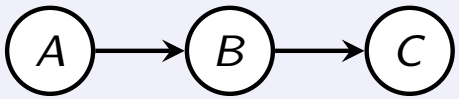
A, C associated?

Age → admitted ← condition (studying only admitted patients)

See Neal (2020), Sections 3.4–3.6.

Three graphical building blocks

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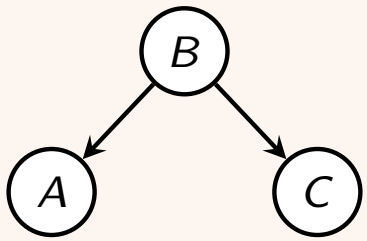
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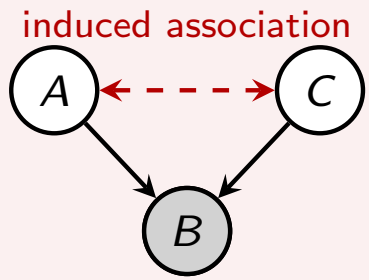
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Yes, *B* is a common cause

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Collider



A, *C* **associated?**

No, *B* blocks

Condition on *B*: **opens**!

Age → admitted ← condition (studying only admitted patients)

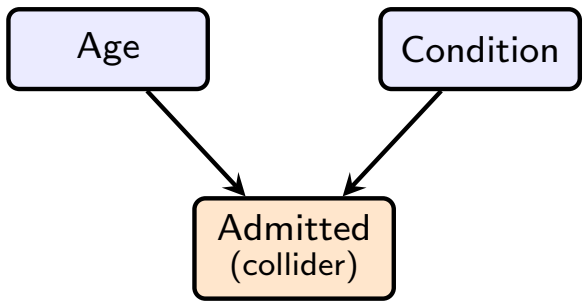
See Neal (2020), Sections 3.4–3.6.

Discuss: collider bias in hospital admissions

Discuss with your neighbour (1 min)

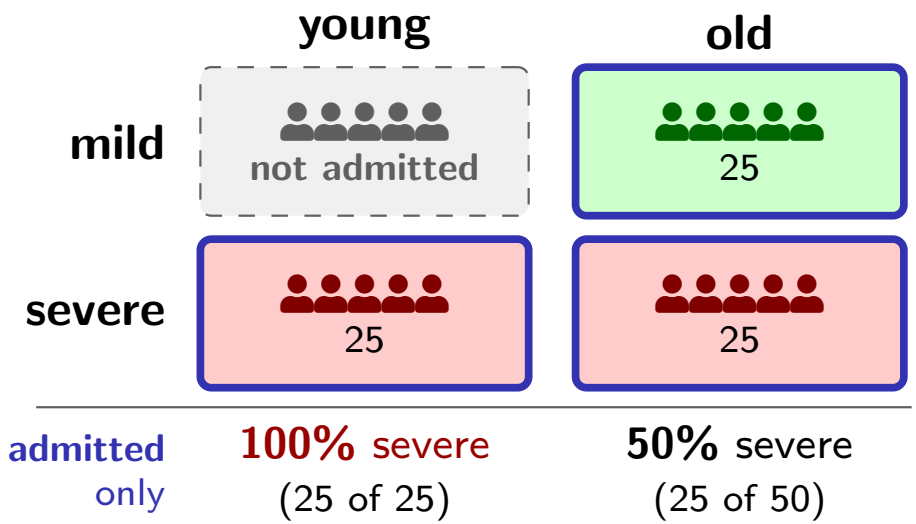
A hospital studies COVID-X using *only admitted patients*. Patients are admitted if they are **old** *or* their condition is **severe**. In the whole population, age and condition are independent.

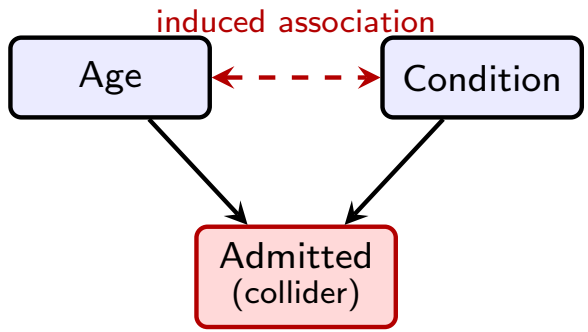
Among *admitted* patients, are older patients more or less likely to be severe than younger ones?



See Neal (2020), Section 3.6 (Berkson's paradox).

Discuss: collider bias in hospital admissions





Illustrative numbers

- Young patients are admitted *only* if severe, so among admitted patients you find: *"older patients have milder COVID-X"*.
- This is an **artefact of selection** (Berkson's paradox).

$\Rightarrow$ Conditioning on a collider (or any of its descendants) opens a non-causal path between its parents.

See Neal (2020), Section 3.6 (Berkson's paradox).

Part 4

The do-operator

Observing is not the same as acting

Potential outcomes | Assumptions | Causal graphs | **do-operator** | Identification

Pearl's do-operator

Observing

$P(Y \mid T = 1)$



patients who *happened* to get Treatment 1 (mostly severe)

↓

Death rate Y

Includes confounding: *why* they got Treatment 1 (severity) is baked in

Intervening

$P(Y \mid do(T = 1))$



we *give* Treatment 1 to *every* patient, mild or severe

↓

Death rate Y

Confounding removed: T is set by us. A causal claim.

$\Rightarrow$ Observing is not the same as intervening.

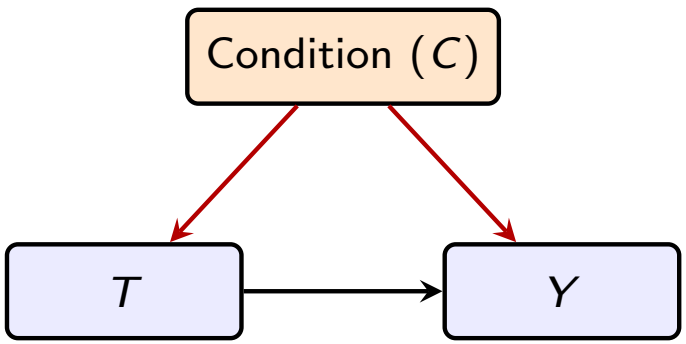
$\Rightarrow$ The two coincide when there is **no confounding** (no variable affects both T and Y), e.g. when T is randomised.

See Neal (2020), Section 4.1.

Intervention as graph surgery

- $do(T = t)$: **remove all incoming edges** to T, then set $T = t$. For COVID-X: *give* treatment t regardless of the patient's condition.

Before $do(T = t)$

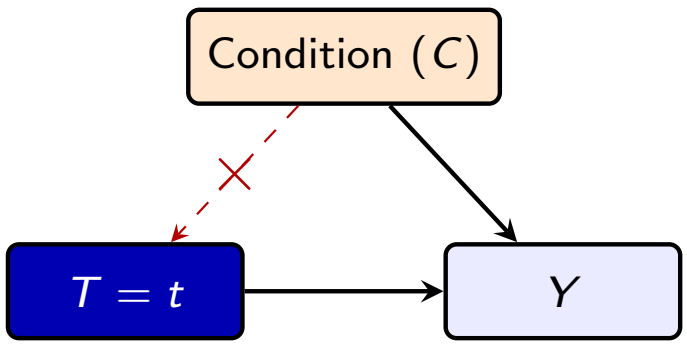


C confounds T and Y

$$\mathbb{E}[Y \mid T = t] \neq \mathbb{E}[Y(t)]$$

$do(T = t)$

After $do(T = t)$



edge $C \to T$ deleted; T set to t by us

$$\mathbb{E}[Y \mid do(T = t)] = \mathbb{E}[Y(t)]$$

$\Rightarrow$ No confounding path from T to Y through C remains, so the interventional distribution equals the potential outcome distribution.

See Neal (2020), Sections 4.1–4.2.

Part 5

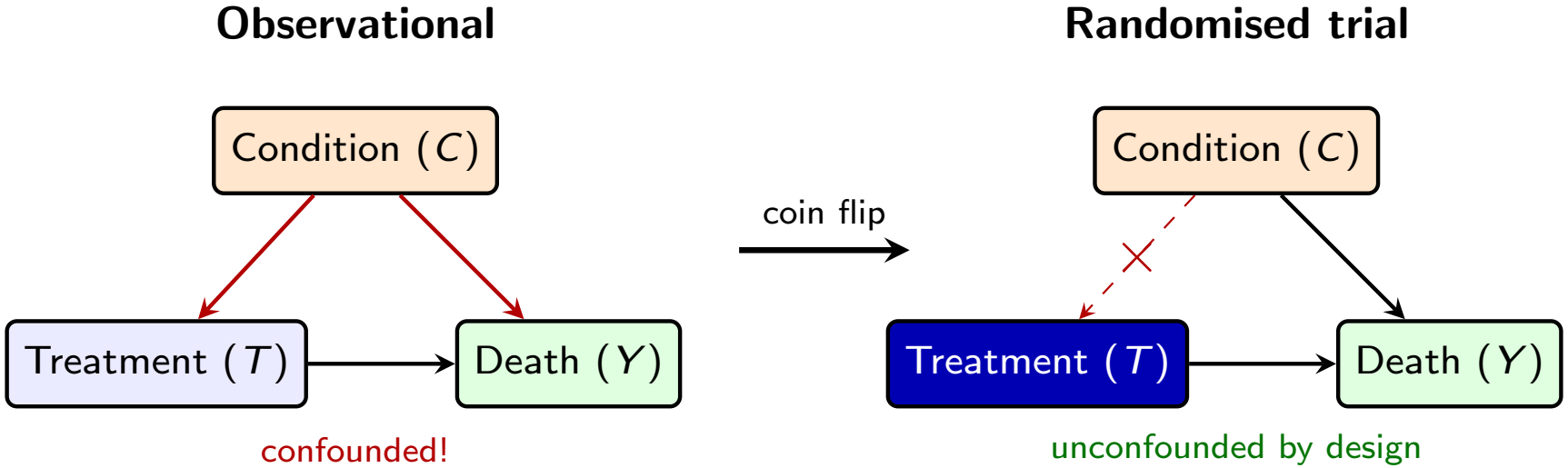
Identification

When and how can we estimate causal effects from data?

Potential outcomes | Assumptions | Causal graphs | do-operator | **Identification**

Randomised experiments: the gold standard

- Randomisation performs graph surgery **automatically**, without us having to intervene on the system.



- A coin flip decides Treatment 0 or 1: both groups get the same mix of mild and severe patients, so Simpson's paradox cannot arise.

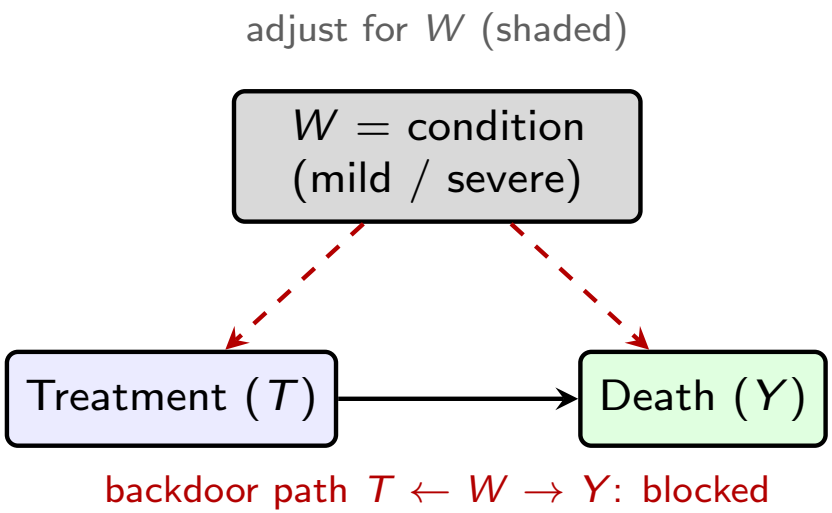
$\Rightarrow$ T is independent of everything that would naturally cause it: all three assumptions hold **by construction**.

See Neal (2020), Chapter 5.

The backdoor adjustment

When we can't randomise, but can observe the right covariates W:

$$\mathbb{E}[Y(t)] = \sum_{w} \mathbb{E}[Y \mid T = t, W = w]\, P(W = w)$$



Backdoor criterion: W must

1. block all backdoor paths from T to Y;
2. contain no descendant of T.

Recipe

- Fit $\mathbb{E}[Y \mid T = t, W = w]$ from data
- Weight by $P(W = w)$
- Recover the ATE: no randomised trial needed

$\Rightarrow$ Observational data *can* identify causal effects, but only under the backdoor criterion and the three identification assumptions.

See Neal (2020), Section 4.4.

Back to COVID-X: adjusting for condition

Condition C blocks the backdoor path $T \leftarrow C \rightarrow Y$, so adjust for it.

Overall, $P(\text{mild}) = \frac{1450}{2050} \approx 0.71$, $P(\text{severe}) = \frac{600}{2050} \approx 0.29$.

Death rate	Mild	Severe	Total
T=0	15%	30%	16%
T=1	10%	20%	19%

patients: 1450 mild, 600 severe

Treatment	Naive $P(\text{die} \mid t)$	Adjusted $\sum_c P(\text{die} \mid t, c)\, P(c)$
Treatment 0	16%	$0.15 \times 0.71 + 0.30 \times 0.29 \approx 19.4\%$
Treatment 1	19%	$0.10 \times 0.71 + 0.20 \times 0.29 \approx 12.9\%$

- Treatment 1 saves lives: about 6.5 fewer deaths per 100 patients than Treatment 0.

$\Rightarrow$ The naive ranking was **backwards**. With the causal structure, Simpson's paradox is not a paradox at all.

Adjusted rates computed from Neal (2020) Table 1.1 with the backdoor adjustment of Section 4.4.

Summary

- **Prediction $\neq$ causation**: a model that predicts perfectly can still give wrong causal claims; the gap is structural, not about data size
- **Potential outcomes** $Y(t)$: only one is ever observed per unit, so causal inference is a missing data problem
- **Three assumptions**: unconfoundedness, positivity, SUTVA
- **Graphs**: forks create confounding; colliders create bias when conditioned on; $do(\cdot)$ is not conditioning
- **Identification**: randomised experiments guarantee it; observational data can too, via the backdoor adjustment, given the right covariates

Reading

- Neal, B., *Introduction to Causal Inference from a Machine Learning Perspective* (2020). Free online. Chapters 1–5.
- Holland, P. W., "Statistics and Causal Inference", *Journal of the American Statistical Association* 81(396), 945–960 (1986).